Cerebellum

It is second largest part of the brain formed of two large sized hemispheres and middle vermis. It coordinates voluntary movements and also maintains body balance.

Pons

'*Pons*' a latin word meaning bridge. It is a bridge of nerve fibre that connects the lobes of cerebellum. It relay signals between the cerebellum, spinal cord, midbrain and cerebrum. It controls respiration and sleep cycle.

Medulla Oblongata

Medulla oblongata is the posterior most part of the brain that connects spinal cord and various parts of brain. It has cardiac centres, respiratory centres, vasomotor centres to control heart beat, respiration and contractions of blood vessels respectively. It also regulates vomiting and salivation.

Table 15.1 Overview of brain functions

Structure	Functions
Cerebral cortex	Sensory preception, control of voulntary functions, language, thinking, memory, decision making, creativity
Thalamus	Acts as relay station
Hypothalamus	Temperature control, thirst, hunger, urination, important link between nervous system and endocrine glands
Cerebellum	Maintenance of posture and balance,coordinate voluntary muscle activity
Pons and medulla	Role in sleep-awake cycle, cardiovascular, respiratory and digestive control centers

More to Know

Electroencephalogram (EEG) is an instrument which records the electrical impulses of brain. An EEG can detect abnormalities in the brain waves and help in diagnoses of seizures, epilepsy, brain tumors, head injuries,etc.

Spinal Cord

Spinal cord is a cylindrical structure lying in the neural canal of the vertebral column. It is also covered by meninges. It extends from the lower end of medulla oblongata to the first lumbar vertebra. The posterior most region of spinal cord tapers into a thin fibrous thread like structure called **filum terminale.**

Internally, the spinal cord contains a cerebrospinal fluid filled cavity known as the **central canal**. The grey matter of spinal cord is 'H' shaped. The upper end of letter 'H" forms **posterior horns** and lower end forms **anterior horns**. A bundle of fibres pass into the posterior horn forming **dorsal** or **afferent root**. Fibres pass outward from the anterior horn forming **ventral** or **efferent root**. These two roots joins to form **spinal nerves**. The white matter is external and have bundle of nerve tracts. Spinal cord conducts sensory and motor impulses to and from the brain. It controls reflex actions of the body.

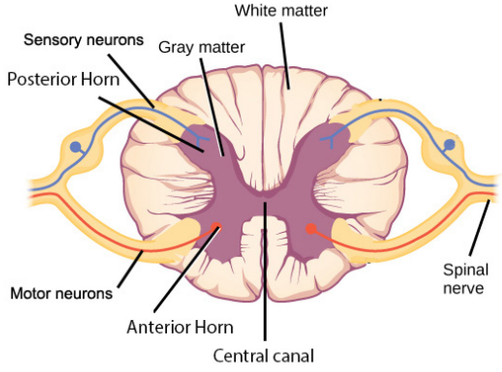


Fig. 15.6 Structure of spinal cord